

# Some Negative Loglikelihood Components for Fitting Composition Data

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Here we describe some negative loglikelihood (NLL) components for fitting composition data for an integrated assessment model. The goal is to provide analytical support for composition model selection in an integrated stock assessment. We begin with standard terminology to describe the data and parameters needed for the statistical models to fit composition data (Table 1).

Table 1. Glossary of terms.

| Variable      | Description                                                                    |
|---------------|--------------------------------------------------------------------------------|
| $C_{max}$     | Number of categories of age or size bins indexed by $c = 1, 2, \dots, C_{max}$ |
| $p_{y,c}$     | Observed proportion for category $c$ in year $y$                               |
| $n_{input,y}$ | Assigned sample size for the observed proportions in year $y$                  |
| $\pi_{y,c}$   | Predicted proportion for category $c$ in year $y$                              |
| $\theta$      | Weighting parameter for the linear Dirichlet-multinomial distribution          |
| $\beta$       | Weighting parameter for the saturating Dirichlet-multinomial distribution      |
| $n_{eff,y}$   | Effective sample size for the observed proportions in year $y$                 |

Here is the multinomial NLL for a single time period, where underlining indicates a vector with dimension number of categories. The vector of estimated, or predicted proportions  $\underline{\pi} = (\pi_1, \dots, \pi_C)$ , is a function of the input sample size  $n_{input}$  and the observed proportions  $\underline{p} = (p_1, \dots, p_C)$ .

$$(1.1) \quad NLL_{MN}(\underline{\pi} | n_{input}, \underline{p}) = -n_{input} \sum_c p_c \cdot \ln(\pi_c)$$

Here is the multinomial NLL for a number of time periods indexed by  $y$ , where double underlining indicates a matrix with dimensions number of categories and number of time periods.

$$(1.2) \quad NLL_{MN}(\underline{\underline{\pi}} | \underline{\underline{n_{input}}}, \underline{\underline{p}}) = -\sum_y n_{input,y} \sum_c p_{y,c} \cdot \ln(\pi_{y,c})$$

Here is the linear Dirichlet Multinomial NLL for a single time period. The vector of predicted proportions  $\underline{\pi} = (\pi_1, \dots, \pi_C)$  and the predicted linear scale parameter  $\theta$  are a function of the input sample size  $n_{input}$  and the observed proportions  $\underline{p} = (p_1, \dots, p_C)$ .

$$(1.3) \quad \begin{aligned} NLL_{DML}(\underline{\pi}, \theta | n_{input}, \underline{p}) = & -\ln(\Gamma(n_{input} + 1)) + \sum_c \ln(\Gamma(n_{input} p_c + 1)) \\ & -\ln(\Gamma(\theta n_{input})) + \ln(\Gamma(n_{input} + \theta n_{input})) \\ & -\sum_c \ln(\Gamma(n_{input} p_c + \theta n_{input} \pi_c)) + \sum_c \ln(\Gamma(\theta n_{input} \pi_c)) \end{aligned}$$

Here is the linear Dirichlet Multinomial NLL for a set of time periods indexed by  $y$ .

$$(1.4) \quad NLL_{DML}(\underline{\underline{\pi}}, \theta | \underline{\underline{n_{input}}}, \underline{\underline{p}}) = \sum_y \left\{ \begin{aligned} & -\ln(\Gamma(n_{input,y} + 1)) + \sum_c \ln(\Gamma(n_{input,y} p_{y,c} + 1)) \\ & -\ln(\Gamma(n_{input,y} \theta)) + \ln(\Gamma(n_{input,y} \theta + n_{input,y})) \\ & -\sum_c \ln(\Gamma(n_{input,y} p_{y,c} + \theta n_{input,y} \pi_{y,c})) + \sum_c \ln(\Gamma(\theta n_{input,y} \pi_{y,c})) \end{aligned} \right\}$$

Here is the saturating Dirichlet Multinomial NLL for a single time period. The vector of predicted proportions  $\underline{\pi} = (\pi_1, \dots, \pi_C)$  and the predicted saturation parameter  $\beta$  are a function of the input sample size  $n_{input}$  and the observed proportions  $\underline{p} = (p_1, \dots, p_C)$ .

$$\begin{aligned}
(1.5) \quad NLL_{DMS}(\underline{\pi}, \beta | n_{input}, \underline{p}) = & -\ln(\Gamma(n_{input} + 1)) + \sum_c \ln(\Gamma(n_{input} p_c + 1)) \\
& -\ln(\Gamma(\beta)) + \ln(\Gamma(\beta + n_{input})) \\
& -\sum_c \ln(\Gamma(n_{input} p_c + \beta \pi_c)) + \sum_c \ln(\Gamma(\beta \pi_c))
\end{aligned}$$

Here is the saturating Dirichlet Multinomial NLL for a set of time periods indexed by  $y$ .

$$(1.6) \quad NLL_{DMS}(\underline{\pi}, \beta | \underline{n_{input}}, \underline{p}) = \sum_y \left\{ \begin{aligned} & -\ln(\Gamma(n_{input,y} + 1)) + \sum_c \ln(\Gamma(n_{input,y} p_{y,c} + 1)) \\ & -\ln(\Gamma(\beta)) + \ln(\Gamma(\beta + n_{input,y})) \\ & -\sum_c \ln(\Gamma(n_{input,y} p_{y,c} + \beta \pi_{y,c})) + \sum_c \ln(\Gamma(\beta \pi_{y,c})) \end{aligned} \right\}$$

The effective sample size  $n_{eff}$  for the linear Dirichlet Multinomial in a single period depends on  $\theta$  and  $n_{input}$  as

$$(1.7) \quad n_{eff} = \frac{1 + \theta n_{input}}{1 + \theta}$$

And for a set of periods  $n_{eff}$  is

$$(1.8) \quad n_{eff} = \sum_y n_{eff,y} = \sum_y \frac{1 + \theta n_{input,y}}{1 + \theta}$$

Similarly, the effective sample size  $n_{eff}$  for the saturated Dirichlet Multinomial in a single period depends on  $\beta$  and  $n_{input}$  as

$$(1.9) \quad n_{eff} = \frac{n_{input} + \beta n_{input}}{n_{input} + \beta}$$

And for a set of periods  $n_{eff}$  is

$$(1.10) \quad n_{eff} = \sum_y n_{eff,y} = \sum_y \frac{n_{input,y} + \beta n_{input,y}}{n_{input,y} + \beta}$$

First order partial derivatives for the effective sample size  $n_{eff}$  of the linear Dirichlet Multinomial as a function of the input parameters are:

$$(1.11) \quad \frac{\partial n_{eff}}{\partial n_{input}} = \frac{\theta}{1 + \theta} = \frac{\theta}{1 + \theta n_{input}} n_{eff}$$

And

$$(1.12) \quad \frac{\partial n_{eff}}{\partial \theta} = \frac{n_{input} - 1}{(1 + \theta)^2} = \frac{n_{input} - 1}{(1 + \theta)(1 + \theta n_{input})} n_{eff}$$

Similarly, first order partial derivatives for the effective sample size  $n_{eff}$  of the saturated Dirichlet Multinomial as a function of the input parameters are:

$$(1.13) \quad \frac{\partial n_{eff}}{\partial n_{input}} = \frac{\beta(1 + \beta)}{(n_{input} + \beta)^2} = \frac{\beta}{n_{input}(n_{input} + \beta)} n_{eff}$$

And

$$(1.14) \quad \frac{\partial n_{eff}}{\partial \beta} = \frac{n_{input}(n_{input} - 1)}{(n_{input} + \beta)^2} = \frac{(n_{input} - 1)}{(1 + \beta)(n_{input} + \beta)} n_{eff}$$